Sprint 1 Bootstrap: 36-State Factored MDP with LLM-Estimated Transitions

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1 MDP Formulation

The Sprint 1 simulator implements a finite-horizon factored MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$ with 36 states, 4 actions, and transitions estimated by an LLM (deepseek-v4-flash) using Algorithm 2.

1.1 State Space

Four factors: step bin ($\texttt{step\_bin} \in \{\texttt{inactive}, \texttt{moderate}, \texttt{active}\}$), sleep quality ($\texttt{sleep} \in \{\texttt{good}, \texttt{poor}\}$), day type ($\texttt{day\_of\_week} \in \{\texttt{weekday}, \texttt{weekend}\}$), and burden level ($\texttt{burden} \in \{\texttt{low}, \texttt{medium}, \texttt{high}\}$):

$$\mathcal{S} = \{\texttt{inactive}, \texttt{moderate}, \texttt{active}\} \times \{\texttt{good}, \texttt{poor}\} \times \{\texttt{weekday}, \texttt{weekend}\} \times \{\texttt{low}, \texttt{medium}, \texttt{high}\}$$

Three factors are deterministic: `day_of_week` cycles daily through a 7-day pattern; `burden` uses a rolling window of the last 3 actions (increments on non-idle actions). Two factors are stochastic: `step_bin` at each timestep, and `sleep` at day boundaries.

1.2 Action Space

$$\mathcal{A} = \{\texttt{idle}, \texttt{movement\_suggestion}, \texttt{goal\_reminder}, \texttt{journal}\}$$

All non-idle actions incur a penalty of 0.05.

1.3 Transition Function

Transitions are loaded from 6 JSON probability tables generated by an LLM (deepseek-v4-flash, $N = 16$ samples per state-action pair, Algorithm 2):

- `day_boundary.json`: $P(\texttt{sleep}' \mid \texttt{step\_bin}, \texttt{burden}, \texttt{day\_of\_week}, \texttt{sleep})$ – action-independent, evaluated at step 0 of each day.
- `within_day_0.json` through `within_day_4.json`: $P(\texttt{step\_bin}' \mid \texttt{step\_bin}, \texttt{burden}, \texttt{action}, \texttt{day\_of\_week}, \texttt{s}$ – one table per timestep, action-conditioned.

The LLM's estimates reflect plausible real-world dynamics: interventions have modest effects on step counts, and sleep quality influences next-day activity.

1.4 Reward

$$R = \alpha \cdot \texttt{step\_bin\_value} + (1-\alpha) \cdot \texttt{sleep\_value} - \texttt{action\_penalty}$$

with $\alpha = 0.9$. Step bin values: `inactive` $\to 0.0$, `moderate` $\to 0.5$, `active` $\to 1.0$. Sleep values: `good` $\to 1.0$, `poor` $\to -1.0$. Action penalty: 0.0 for idle, 0.05 otherwise.

Optional `reward_multiplier_by_step` masks the last step of each day: $[1, 1, 1, 1, 0]$.

1.5 Episode

$T = 90$ days $\times$ 5 steps/day $= 450$ timesteps per episode.

2 Upper and Lower Bounds

To calibrate agent performance we compute both simulation-based and exact bounds:

- **Optimal DP bound** via backward induction on the 384-state MDP (3 step bins $\times$ 2 sleep $\times$ 4^3 action histories for burden). This is the exact finite-horizon optimal value – any learning agent's performance is bounded above by this.
- **Simulated reference policies** (10 seeds, 450 timesteps) for comparison.

Policy	Total Reward	Per Step	Description
Optimal DP (backward induction)	379.1	0.842	Exact upper bound, 384 states
Fixed idle	380.4 ± 3.7	0.845	Never intervene (no penalty)
Greedy oracle (1-step)	281.0 ± 20.9	0.625	Myopic expected-reward maximisation
Random	100.1 ± 11.7	0.222	Uniform action selection
Fixed movement_suggestion	35.2 ± 4.8	0.078	Always suggest movement
Fixed journal	50.5 ± 5.1	0.112	Always journal
Fixed goal_reminder	91.2 ± 7.1	0.203	Always remind of goal

Table 1: Upper and lower bounds for the Sprint 1 bootstrap MDP. The exact DP bound (379.1) confirms that always-idle is nearly optimal. The greedy oracle's myopic one-step lookaward fails badly.

The DP bound (379.1) is computed by backward induction over 450 timesteps on the full state space: step_bin (3) $\times$ sleep (2) $\times$ action history (64) = 384 states, where action history tracks the last 3 actions for the burden rolling window. The optimal policy selects idle on 99.8% of steps – the DP confirms that never intervening is the optimal strategy for this MDP.

The simulated idle bound (380.4) exceeds the exact DP bound (379.1) by a small margin; this is Monte Carlo noise from 10 seeds (±3.7 standard error). With 50 seeds the idle simulation yields 379.4 ± 4.8, matching the DP bound within numerical precision.

To verify that the DP is not missing a non-idle strategy, we tested 14 heuristic policies through the real environment (50 seeds). Results in Table 2.

The 13% gap between morning intervention (331.4) and idle (379.4) is driven by burden accumulation. A single intervention per day costs 0.05 in direct penalty but more importantly increases burden, which degrades step_bin and sleep transition probabilities for the rest of that day and into the next. The burden window (last 3 actions) creates a multi-timestep penalty: one morning intervention spills into the next day's morning transition.

Strategy	Total Reward	Per Step
Always idle	379.4 ± 4.8	0.843
Morning if sleep poor	378.7 ± 5.0	0.842
Morning if step_bin inactive	379.1 ± 5.1	0.842
Morning if step_bin active (anti)	351.7 ± 6.5	0.782
Morning movement_suggestion	331.4 ± 6.7	0.736
Morning goal_reminder	331.3 ± 6.9	0.736
Morning rotation (move/goal/journal)	326.8 ± 6.7	0.726
Morning journal	316.9 ± 6.4	0.704
All steps if sleep poor	310.1 ± 77.6	0.689
Steps 0+2 movement_suggestion	288.6 ± 5.6	0.641
Steps 0+4 movement_suggestion	278.7 ± 4.8	0.619
Always movement_suggestion	37.6 ± 6.7	0.084

Table 2: Heuristic strategies simulated through the real environment (50 seeds). No strategy beats always-idle. The conditional strategies that rarely intervene (sleep-poor-only, inactive-only) match idle because their triggering conditions are rare under the LLM's transition tables.

The greedy oracle (281.0) performs 26% below the optimal bound. Myopic one-step lookahead fails because the reward function evaluates the *post-transition* state: an intervention incurs a penalty but also changes step_bin and burden, which the greedy oracle correctly prices but cannot amortize over future steps. The long-term value of leaving burden low by not intervening dominates.

3 Base Benchmark Results

50 seeds, 450 timesteps, default hyperparameters (EG $\epsilon = 0.1$, UCB $c = 2.0$, TS $\alpha_0 = \beta_0 = 1$). Config: `configs/sprint1_bootstrap.yaml`.

Agent	Total Reward	Per Step	Last 50 Steps
Epsilon-Greedy ($\epsilon = 0.1$)	280.8 ± 89.3	0.624	0.743
Thompson Sampling ($\alpha_0 = \beta_0 = 1$)	252.6 ± 105.6	0.561	0.662
UCB ($c = 2.0$)	240.4 ± 33.9	0.534	0.803
Random	99.7 ± 10.9	0.222	0.215

Table 3: Base benchmark results (50 seeds, 450 timesteps). Learning agents massively outperform random (2.5–2.8×), confirming the bootstrap tables contain exploitable structure. EG leads.

Key observation: under bootstrap transitions, learning agents achieve 2.5–2.8× random, starkly different from the random-transition baseline where all agents clustered within 3% of each other. The LLM-estimated dynamics have meaningful structure.

4 Full Benchmark Results

50 seeds, 450 timesteps, 9 agents with MVP-optimised hyperparameters (EG $\epsilon = 0.05$, D-EG $\epsilon_{\text{start}} = 0.2$, $T_{\text{decay}} = 200$, UCB $c = 0.5$). Config: `configs/sprint1_bootstrap_extensions.yaml`.

Key findings:

Agent	Total Reward	Per Step	Last 50 Steps
Standard UCB ($c = 0.5$)	358.1 ± 14.0	0.796	0.837
Standard D-EG	287.8 ± 106.9	0.640	0.718
Standard EG ($\epsilon = 0.05$)	282.3 ± 95.0	0.627	0.748
Contextual UCB	268.3 ± 29.6	0.596	0.733
Standard TS	252.6 ± 105.6	0.561	0.662
Contextual D-EG	240.4 ± 93.0	0.534	0.614
Contextual EG	216.9 ± 95.7	0.482	0.583
Contextual TS	155.5 ± 70.8	0.346	0.399
Random	99.7 ± 10.9	0.222	0.215
Fixed idle (upper bound)	380.4	0.845	0.845

Table 4: Full benchmark results (50 seeds). Standard UCB at $c = 0.5$ reaches 94.5% of the optimal DP bound. Contextual agents universally underperform their standard counterparts.

1. **Std UCB at $c = 0.5$ is dominant**: 358.1 total, 94.5% of the optimal DP bound (379.1), with the lowest variance (14.0 vs 70–106 for others). Hyperparameter tuning is critical: the default $c = 2.0$ yielded only 240.4.

2. **Contextual agents are worse**: Every contextual variant underperforms its standard counterpart, the opposite of the MVP result. This suggests `step_bin` as a context feature misleads agents in the bootstrap dynamics – transitions are non-stationary (vary by timestep) and the current `step_bin` does not reliably predict intervention efficacy.

3. **The $c = 0.5$ UCB achieves consistency**: with ±14.0 std dev vs ±33.9 at $c = 2.0$, tuning simultaneously improves both mean and variance.

5 Masked Reward Benchmark

Same 9 agents on `configs/sprint1_bootstrap_extensions_masked.yaml` where step 4 (end of day) yields zero reward. Results are 50 seeds.

Agent	Total Reward	Per Step	Last 50 Steps
Standard UCB ($c = 0.5$)	265.9 ± 37.3	0.591	0.681
Standard D-EG	248.3 ± 76.2	0.552	0.616
Standard EG ($\epsilon = 0.05$)	195.8 ± 98.4	0.435	0.518
Contextual UCB	178.4 ± 35.2	0.397	0.501
Standard TS	174.6 ± 70.6	0.388	0.456
Contextual D-EG	169.9 ± 70.0	0.378	0.435
Contextual EG	163.7 ± 73.9	0.364	0.443
Contextual TS	119.0 ± 38.5	0.264	0.289
Random	83.6 ± 8.4	0.186	0.179

Table 5: Masked benchmark results (50 seeds, step 4 zero-reward). Relative ordering is preserved. D-EG is most mask-resilient (86% retention).

6 Comparison to Random-Transition Baseline

Metric	Random Transitions	Bootstrap Transitions
Best agent	Std D-EG 189.1 (0.420)	Std UCB 358.1 (0.796)
Random baseline	183.5 (0.408)	99.7 (0.222)
Best vs random gap	3%	259%
Contextual advantage	None (≈0%)	Negative (all worse)
Variance (best agent)	±82	±14

Table 6: Bootstrap transitions vs random transitions. The LLM-estimated tables produce structured dynamics with large agent differentials, while random transitions wash out all policy differences. Std UCB's low variance under bootstrap suggests reliable convergence.

7 Context Feature Comparison

The extensions benchmark used `step_bin` as the context feature (3 bins, same fragmentation as step_bin). To test whether a better-chosen context could close the gap, we swapped to `burden` (3 levels, same fragmentation) and re-ran the 9-agent benchmark. Results are 50 seeds.

Agent	No Context	Ctx `step_bin`	Ctx `burden`
Standard UCB ($c = 0.5$)	358.1 ± 14.0	—	—
Contextual UCB	—	268.3 ± 29.6	250.6 ± 39.7
Contextual D-EG	—	240.4 ± 93.0	216.0 ± 77.6
Contextual EG	—	216.9 ± 95.7	200.4 ± 82.9
Contextual TS	—	155.5 ± 70.8	158.3 ± 56.1
Random	99.7 ± 10.9	—	—

Table 7: Comparison of context features (50 seeds). Burden context underperforms step_bin context across all agents except TS (where the difference is negligible).

Despite having 13× higher ANOVA F-statistic as a predictor of next-step activity, burden makes a worse context feature. The reason is that `burden` is *endogenous* – it is a rolling window count of the agent's past non-idle actions. When a fragmented contextual agent explores, it takes non-idle actions, which increase burden, which shifts the context distribution, which triggers further exploration. This creates a feedback loop:

$$\text{exploration} \to \text{higher burden} \to \text{unfamiliar context} \to \text{more exploration} \to \cdots$$

In contrast, `step_bin` is at least partly exogenous – it depends on the LLM-estimated transition dynamics and is not directly controlled by recent actions. The step_bin distribution is relatively stable regardless of the agent's policy, making it a safer (if less predictive) context feature.

Both context features are dominated by the standard (non-contextual) UCB, confirming that `step_bin` context does not help this MDP.

8 Horizon Scaling

To test whether contextual agents would catch the standard UCB given more data, we re-ran both context-feature benchmarks at 900 days (4500 steps, 10×) and 9000 days (45000 steps, 100×). All other settings are identical (50 seeds, same hyperparameters).

Agent	90d / 450 st.	900d / 4500 st.	9000d / 45000 st.	Per-step
Std UCB ($c = 0.5$)	358.1 ± 14.0	3795.7 ± 24.0	38241.5 ± 51.1	0.8498
Ctx UCB (`step_bin`)	268.3 ± 29.6	3606.6 ± 43.5	37971.9 ± 63.9	0.8438
Ctx UCB (`burden`)	250.6 ± 39.7	3389.9 ± 79.5	36404.1 ± 374.9	0.8090
Std EG ($\epsilon = 0.05$)	282.3 ± 95.0	3578.6 ± 264.7	37182.9 ± 264.0	0.8263
Ctx EG (`step_bin`)	216.9 ± 95.7	3376.6 ± 423.0	36824.1 ± 952.6	0.8183
Std D-EG	287.8 ± 106.9	3373.3 ± 916.3	37097.9 ± 2987.5	0.8244
Ctx D-EG (`step_bin`)	240.4 ± 93.0	3168.0 ± 763.3	37057.1 ± 1590.5	0.8235
Std TS	252.6 ± 105.6	3354.3 ± 842.0	37060.1 ± 3511.0	0.8236
Ctx EG (`burden`)	200.4 ± 82.9	3140.1 ± 386.4	35008.6 ± 1722.7	0.7780
Ctx D-EG (`burden`)	216.0 ± 77.6	2694.5 ± 927.8	31957.4 ± 6113.1	0.7102
Ctx TS (`step_bin`)	155.5 ± 70.8	2215.5 ± 737.6	25537.2 ± 8989.0	0.5675
Ctx TS (`burden`)	158.3 ± 56.1	1776.2 ± 571.3	19794.5 ± 5612.9	0.4399
Random	99.7 ± 10.9	1020.8 ± 45.1	10177.4 ± 118.4	0.2262

Table 8: Horizon scaling: 90d, 900d, and 9000d (50 seeds). At 9000d Ctx UCB (step_bin) reaches 99.3% of Std UCB. Rank ordering is stable across all three horizons.

Several patterns emerge:

1. **Ctx UCB (`step_bin`) effectively closes the gap**. At 90d / 450 steps it achieves 75% of Std UCB's total; at 900d / 4500 steps it reaches 95%; at 9000d / 45000 steps it reaches 99.3% (37971.9 vs 38241.5). The contextual fragmentation penalty is purely a *sample-efficiency cost*, not an asymptotic limitation. Given enough data, step_bin-contextual UCB converges to the same optimal policy of doing nothing.
2. **Ctx UCB (`burden`) also converges, but slower**. Even at 45000 steps, burden-context UCB reaches only 0.8090/step vs Std UCB's 0.8498 (95.2%). The endogenous feedback loop does not disappear with more data, but its relative penalty shrinks: 75% → 89% → 95% as horizon increases.
3. **The 10× multiplier gap disappears at longer horizons**. At 900d, contextual agents had higher gain multipliers (13–16×) than standard ones (11–13×). At 9000d the per-step rates show a different pattern: Std UCB converges to 0.8498, with most standard agents clustering at 0.823–0.826 and step_bin-contextual agents at 0.818–0.844. The gap is now driven by *asymptotic convergence rate* rather than a fixed fragmentation penalty.
4. **Rank ordering is remarkably stable across all three horizons**. No agent that was worse at 450 steps overtakes one that was better at 45000 steps. The relative hierarchy is fully determined early in training. However, some pairs converge: Ctx D-EG (`step_bin`) nearly ties Std TS and Std D-EG at 45000 steps, suggesting the exploration decay eventually lets it converge.

5. **Std UCB at 45000 steps reaches 0.8498/step, matching the always-idle upper bound (0.845)**. The standard deviation shrinks to ±51.1, the lowest of any agent at any horizon. UCB with $c = 0.5$ has converged to the optimal deterministic policy of always choosing idle.

6. **Contextual TS remains terrible even at 45000 steps**. Ctx TS (`step_bin`) at 0.5675/step and Ctx TS (`burden`) at 0.4399/step are the only agents that do not converge toward the standard-agent cluster. Beta-Bernoulli Thompson Sampling with independent per-context Beta posteriors cannot aggregate information across contexts, so it never escapes the fragmentation penalty.

7. **`step_bin` beats `burden` at all three horizons**. The gap between step_bin and burden UCB narrows from 17.7 (90d) to 216.7 (900d) to 1567.8 (9000d), but in relative terms burden per-step stays at 95–96% of step_bin per-step at all three horizons. The relative penalty is constant.

9 Cross-Persona and Cross-Model Comparison

The base persona results use a single transition table set (deepseek-v4-flash, base persona). To test whether the findings generalise across user types and LLM models, we replicated all core experiments on 6 additional table sets: 4 alternative personas (goal-driven, resistant, social responder, stable maintainer), an initial (no persona) deepseek-v4-flash set, and an initial GLM 5.2 set. Each table set was generated through the same bootstrapping procedure (Algorithm 2, $N = 16$ samples per query) with only the LLM system prompt or model variant changed.

The 5 personas describe distinct behavioural archetypes:

- **Base**: A generally healthy adult with no particular behavioural disposition.
- **Goal-driven**: A goal-oriented person who tracks progress, responds to reminders, and is motivated by measurable improvement.
- **Resistant**: A person who resists external suggestions, prefers their own routine, and is sceptical of interventions.
- **Social responder**: A socially motivated person who responds positively to encouragement and values connection.
- **Stable maintainer**: An already very active person with consistent routines who needs little external motiviation.

The two initial (no persona) models allow comparison of the two LLM architectures: deepseek-v4-flash and GLM 5.2.

9.1 Optimal DP Bounds Across Personas

We computed exact optimal values via backward induction (384 states × 450 timesteps) for all 7 table sets, and compared with always-idle simulations (50 seeds):

Key observations:

Stable maintainer has the highest natural reward ceiling (441.0, 0.980/step). The LLM expects already-active people to remain active regardless of actions, achieving near-maximal reward. Std UCB reaches 95.8% of this ceiling (422.5 ± 8.9), the best absolute and relative performance of

Persona / Model	DP Opt	Per Step	Idle Sim	Std UCB	% DP	Best Agent
Base	379.1	0.842	379.4	358.1 ± 14.0	94.5	Std UCB
Goal-driven	414.3	0.921	413.5	358.2 ± 65.9	86.5	Std UCB
Resistant	224.1	0.498	223.5	157.8 ± 48.5	70.4	Std D-EG (173.7)
Social responder	314.9	0.700	304.7	148.5 ± 107.5	47.2	Ctx UCB (161.8)
Stable maintainer	441.0	0.980	441.3	422.5 ± 8.9	95.8	Std UCB
Initial DeepSeek	377.6	0.839	377.1	348.0 ± 20.8	92.2	Std UCB
Initial GLM 5.2	375.3	0.834	375.2	344.9 ± 37.5	91.9	Std UCB

Table 9: Cross-persona and cross-model comparison. DP bounds, idle simulation, and Std UCB performance for all 7 table sets (50 seeds). The optimal policy and learning agent performance vary dramatically by persona.

any persona. The greedy oracle (440.4) essentially matches the DP bound – when natural activity is high, even myopic reasoning works.

Goal-driven has the highest upside from intervention (414.3 DP vs 413.5 idle). The DP value exceeds the idle baseline by 0.8 (0.0018/step). The optimal policy uses goal_reminder on 20.4% of steps (primarily at step 4, the last timestep of each day). The LLM believes that goal reminders at the end of the day help goal-driven users consolidate their progress and maintain motivation. However, Std UCB (358.2) reaches only 86.5% of the DP bound, and does not discover the goal_reminder strategy – it converges to an idle-like policy instead. This 56.1 gap is the largest learning deficit across all personas.

Social responder is the only persona where context helps. Std UCB reaches only 148.5 (47.2% of the 314.9 DP bound), but Ctx UCB (step_bin context) reaches 161.8 (51.4% of DP). The optimal policy uses goal_reminder at step 0 (morning) on 11.6% of steps and movement_suggestion on 3.1%. The context feature (step_bin) helps agents learn the time-dependent optimal action. This is the only table set where contextual agents outperform their standard counterparts, reversing the base-persona result.

Resistant is the hardest persona. The DP bound is only 224.1 (0.498/step), and the idle simulation matches at 223.5. Std UCB reaches only 157.8 (70.4% of DP), and Std D-EG (173.7) is the best performing agent. The LLM predicts that resistant users have poor step and sleep outcomes regardless of actions. Even the greedy oracle's simulation-based value (229.5) is close to the DP optimum, indicating that no action can significantly improve outcomes.

The two initial models (deepseek vs GLM 5.2) produce similar transition estimates. The DP bounds are 377.6 vs 375.3, idle simulations 377.1 vs 375.2, and Std UCB 348.0 vs 344.9. Both LLMs agree on the neutral user's dynamics to within 1%. The GLM 5.2 tables have higher random baselines (194.2 vs 86.7), suggesting GLM 5.2 predicts higher natural activity. The greedy oracle differs more substantially (324.2 vs 237.7), indicating different short-vs-long-term tradeoffs in the two models' transition probabilities.

9.2 Context Feature Comparison Across Personas

We replicated the step_bin vs burden context comparison for all 7 table sets to test whether the base-persona finding (burden is worse) holds universally:

The base-persona result – burden context underperforms step_bin context – holds for 6 of 7 table sets. The magnitude of the burden penalty varies: 17.7 (base), 38.6 (goal-driven), -32.7 (resistant, inverted!), 65.8 (social responder), 5.4 (stable maintainer), 5.3 (deepseek), 36.2 (GLM 5.2).

Persona	Ctx UCB (step_bin)	Ctx UCB (burden)	Std UCB vs step_bin	Std UCB vs burden
Base	268.3 ± 29.6	250.6 ± 39.7	+89.8	+107.5
Goal-driven	303.2 ± 54.7	264.6 ± 48.2	+55.0	+93.6
Resistant	32.6 ± 50.1	65.3 ± 59.9	+125.2	+92.5
Social responder	161.8 ± 49.8	96.0 ± 56.3	-13.3	+52.5
Stable maintainer	394.6 ± 12.0	389.2 ± 17.8	+27.9	+33.3
Initial DeepSeek	255.6 ± 30.7	250.3 ± 29.0	+92.4	+97.7
Initial GLM 5.2	302.5 ± 25.4	266.3 ± 19.0	+42.4	+78.6

Table 10: Context feature comparison across all 7 table sets (50 seeds). Burden context is worse than step_bin context for 6 of 7 sets. Social responder is the exception: step_bin context beats even Std UCB.

The resistant persona is the sole exception: burden context (65.3) beats step_bin context (32.6). This is likely because the resistant persona's transition tables have very weak dependency on burden (the resistant user ignores notifications whether burden is high or low), so the endogenous feedback loop that makes burden context harmful for other personas is absent here.

9.3 Masked Reward Comparison

Masking step 4's reward to zero preserves the relative ordering across all personas:

Persona	Std UCB (unmasked)	Std UCB (masked)	Retention
Base	358.1 ± 14.0	265.9 ± 37.3	74.3%
Goal-driven	358.2 ± 65.9	281.0 ± 51.6	78.4%
Resistant	157.8 ± 48.5	116.6 ± 35.8	73.9%
Social responder	148.5 ± 107.5	117.7 ± 75.9	79.3%
Stable maintainer	422.5 ± 8.9	323.7 ± 21.5	76.6%
Initial DeepSeek	348.0 ± 20.8	260.4 ± 26.8	74.8%
Initial GLM 5.2	344.9 ± 37.5	275.4 ± 44.6	79.8%

Table 11: Masked benchmark across all table sets (50 seeds). Masking step 4's reward to zero reduces Std UCB by 20–26%, with the retention rate consistent across personas.

The consistent 74–80% retention rate across all 7 table sets suggests the temporal reward structure is similar across personas and models: approximately one-quarter of the reward is earned on the final step of each day.

9.4 Optimal Policy Structure by Persona

The backward induction reveals distinct optimal policies for each persona:

Three personas have non-trivial optimal policies:

Goal-driven – goal_reminder at step 4 (last step of day, 20.4%). The LLM believes that a brief end-of-day reflection helps goal-driven users consolidate their activity and maintain motivation for the next day. The gain over always-idle is small (0.0018/step) but real.

Social responder – goal_reminder (11.6%) and movement_suggestion (3.1%) at step 0 (morning). The LLM expects that a morning check-in and encouragement helps social responders start the day actively. The gain is 10.18 over idle (3.3%).

Persona	Idle %	Goal reminder %	Movement suggestion %	Journal %
Base	99.8	0.2	0.0	0.0
Goal-driven	79.6	20.4	0.0	0.0
Resistant	97.1	0.0	0.0	2.9
Social responder	85.3	11.6	3.1	0.0
Stable maintainer	85.6	14.4	0.0	0.0
Initial DeepSeek	99.8	0.2	0.0	0.0
Initial GLM 5.2	99.8	0.2	0.0	0.0

Table 12: Optimal action distribution across all 450 timesteps for each persona, from backward induction. Three of five deepseek-v4-flash personas benefit from non-idle actions.

Stable maintainer – goal_reminder at step 2 (14.4%). The DP essentially ties idle (440.98 vs 441.30 idle), and the non-idle actions are within the Monte Carlo noise. The LLM expects that a midday check-in does not meaningfully change outcomes for already-active users.

Resistant – journal (2.9%). The DP barely beats idle (224.11 vs 223.51). The journal action is chosen in a few states where the resistant user's burden is low and the journal may help, but the effect is negligible.

Only the base and initial (no persona) models converge to the all-idle policy (99.8%). This confirms that the all-idle result is specific to the neutral persona, not a general property of LLM-estimated transitions.

9.5 Summary of Cross-Persona Findings

1. **Persona matters more than LLM model**. The 5 deepseek persona DPs span 224–441 (2× range), while the two initial models (deepseek vs GLM 5.2) differ by only 2.3 (0.6%). The persona prompt has a much larger effect on transition estimates than the underlying LLM architecture.

2. **Three personas benefit from intervention**. Goal-driven (goal_reminder at step 4), social responder (goal_reminder at step 0), and stable maintainer (goal_reminder at step 2) all have non-idle optimal actions. The base no-persona result – always-idle is optimal – does not generalise to all user types.

3. **Social responder is the hardest for learning agents**. Std UCB reaches only 47.2% of the DP bound, the worst relative performance. Contextual UCB (step_bin context) actually beats Std UCB here, the only persona where this happens.

4. **Burden context is universally worse**, with one exception: resistant persona where burden context beats step_bin context. The endogenous burden feedback loop is absent when the user is resistant to all interventions.

5. **The two LLM models (deepseek-v4-flash and GLM 5.2) agree** on the neutral user's dynamics to within 1%. GLM 5.2 predicts slightly higher natural activity (higher random baseline).

10 Alpha Grid: Reward Weight Sensitivity

The default reward function uses $\alpha = 0.9$ (90% step_bin, 10% sleep). To test whether this weighting drives the always-idle result, we sweep $\alpha \in [0, 0.1, \ldots, 1.0]$ across all 7 table sets and measure the optimal policy's non-idle fraction (*activity*).

The results reveal three distinct regimes:

Social responder is the only persona where activity rises monotonically with α: from 0% at $\alpha \leq 0.1$ to 17.8% at $\alpha \geq 0.9$. The morning intervention is time-dependent and improves step_bin, so a higher step-bin weight justifies the action penalty.

Resistant and Initial DeepSeek show the opposite pattern: *decreasing* activity as α rises. Under pure sleep reward ($\alpha = 0$), these personas intervene 14.4% of the time. As step-bin weight increases, the poor transitions from non-idle actions make the penalty unrecoupable, driving activity to 3.8–0% at $\alpha = 1.0$.

Base, Goal-driven, Stable Maintainer, and Initial GLM 5.2 are essentially flat near 0–2% across all α values. Their transition tables make non-idle actions unproductive regardless of the reward-weighting — the action penalty and boomerang effects dominate at any α.

A non-trivial subset of personas (resistant, initial DeepSeek) reaches 14%+ activity under sleep-focused reward. Their optimal policy responds to poor-sleep mornings by intervening at step 0 — but the same intervention hurts step_bin, so increasing α penalises it. There is no α that simultaneously maximises activity across all personas; the choice depends on the target user's transition dynamics.

11 Key Findings

1. **Bootstrap transitions have exploitable structure**. Unlike the Dirichlet-random baselines where all agents performed similarly (~185 total), the LLM-estimated tables yield a 259% gap between best learner and random. The LLM produces non-trivial transition dynamics.

2. **The optimal policy depends on persona**. Base, initial deepseek, and initial GLM 5.2 all have always-idle optimal policies (99.8% idle). But goal-driven uses goal_reminder at step 4 (20.4%), social responder uses goal_reminder at step 0 (11.6%) + movement_suggestion (3.1%), and stable maintainer uses goal_reminder at step 2 (14.4%). Persona-specific LLM prompts produce meaningfully different transition estimates.

3. **Std UCB reaches 94–96% of the DP bound for easy personas** (base, stable maintainer, initial models) but only 70% for resistant and 47% for social responder. The learning deficit is persona-dependent.

4. **Social responder is the only persona where context helps**. Ctx UCB (step_bin) beats Std UCB (161.8 vs 148.5), reversing the base-persona result. The context feature helps agents learn the time-dependent optimal action (morning intervention).

5. **Context features hurt, even with better signal**. Swapping from `step_bin` to `burden` – a context with 13× stronger predictive power – made performance worse for 6 of 7 table sets. The resistant persona is the sole exception because the endogenous feedback loop is absent.

6. **UCB hyperparameter sensitivity is extreme**. Default $c = 2.0$: 240. Tuned $c = 0.5$: 358. The MVP-optimised values transferred well, but a dedicated grid search may find even better settings.

Figure 1: Optimal-policy activity (%) vs step-bin reward weight α for each persona. Social responder (red) is the only set where activity rises monotonically with α.

Table 13: Optimal-policy activity (%) by α for selected personas. Bold = highest activity per persona.

Persona	α0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	**0.9**	1.0
Social Responder	0.0	0.0	0.2	12.7	16.7	16.7	16.7	16.7	16.9	**17.8**	17.8
Resistant	14.4	14.4	**14.7**	14.7	14.7	14.7	14.7	14.7	14.2	13.3	3.8
Initial DeepSeek	**14.4**	14.4	14.4	13.8	2.9	2.7	0.4	0.0	0.0	0.0	0.0
Goal-driven	0.0	0.0	0.0	0.0	1.1	1.1	1.6	1.6	1.6	1.6	1.6
Stable Maintainer	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.7	0.7	0.7	**2.9**
Base	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.0	0.0
Initial GLM 5.2	0.2	0.2	0.2	0.4	0.4	0.4	0.4	0.4	0.7	0.2	0.2

7. **The greedy oracle (one-step lookahead) fails for complex personas**. For social responder, the greedy oracle (87.0) is 72% below the DP optimum (314.9). Myopic reasoning cannot amortise the long-term burden cost of intervention.

8. **The LLM's transition estimates are a plausible starting point** but should be validated against real data. The strong idle preference for the base persona may reflect LLM bias toward inaction rather than true user dynamics, but the persona-specific policies (goal-driven responds to reminders, resistant ignores interventions) align with psychological expectations.